

QA/QC Engine — 5-Loan Disposition Run

Purpose: Validate the deterministic QA/QC engine against 5 synthetic closed-loan files, each seeded with known planted defects, to confirm the engine correctly identifies every issue with a full audit trail.

Engine: qc_engine v p0-1.0.0 · **Ruleset:** rs-defect-verification v1 (21 authored Check objects, gated per loan by document presence / property age / loan program) · **Run date:** 2026-07-22

Result: All 5 loans correctly routed to **NEEDS_REVIEW** — this run is a validation of engine correctness, not a real production batch, so a 5-for-5 review rate confirms the engine caught every seeded defect rather than false-clearing a bad loan.

Executive Summary

Loan ID	Program	Checks Applied	Failures	Disposition
2025-0917-001	Conventional	9	2	NEEDS_REVIEW
2025-1004-FHA-002	FHA	13	5	NEEDS_REVIEW
2025-1108-VA-003	VA	13	6	NEEDS_REVIEW
2025-1215-FRD-004	Conventional	11	5	NEEDS_REVIEW
2025-1122-USDA-005	USDA	13	8	NEEDS_REVIEW
Total		59	26	5 / 5 flagged

Each loan runs a different subset of the 21-check ruleset — checks are gated to the loan's actual program (e.g. VA-only checks don't run against a USDA loan) and to whether the relevant document is even present in that loan's file. "Checks Applied" above is the gated count, not all 21.

Loan 01 — 2025-0917-001 (Conventional)

2 of 9 applicable checks failed.

Check	Finding	Source
Large deposit source documented	\$15,000 mobile deposit (08/12) has no source documentation in file	Wells Fargo Consumer Checking Statement, p.1
Appraisal comp distance within guideline	Comparable sale 8.5 mi from subject exceeds the 5-mile urban guideline, no addenda explanation	Form 1004 UAD, Comparable Sales Grid, p.1

Loan 02 — 2025-1004-FHA-002 (FHA)

5 of 13 applicable checks failed; 1 informational data-sync flag.

Check	Finding	Source
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HUD-92900-A Section III certification signed	Borrower Certification signature line is blank	HUD-92900-A Addendum, p.1
Gift funds source documented	No donor bank statement, wire/check copy, or borrower receipt in file	Gift Letter — FHA Down Payment, p.1
Lead-paint completion certification present	Peeling paint on pre-1978 structure flagged "subject to," no Form 442 compliance inspection in file	FHA Appraisal Summary 1004/URAR, p.1
FHA Amendatory Clause present	Amendatory Clause / Real Estate Certification not in file	—
Lead-based paint disclosure present	Pre-1978 property, disclosure not in file	—

Informational only (does not drive disposition): FHA case number on the 1003 (381-9927164) does not match FHA Connection's case number (381-9927614) — a system data-sync issue, not a document defect.

Loan 03 — 2025-1108-VA-003 (VA)

6 of 13 applicable checks failed.

Check	Finding	Source
Lead-paint completion certification present	Peeling paint flagged, no Form 442 completion cert in file	—
ARM Pre-Loan Disclosure present	Missing	—
Termite inspection present	VA requires NPMA-33 termite inspection in NC; not in file	VA Appraisal, VA Form 26-1805/URAR, p.1
Lead-based paint disclosure present	Pre-1978 property, disclosure not in file	—
VA residual income documented	Residual income calculation not documented for borrower's family size/region	—
VA NOV issued on or before closing	Notice of Value dated 3 days after closing — invalid date order	—

Loan 04 — 2025-1215-FRD-004 (Conventional)

5 of 11 applicable checks failed.

Check	Finding	Source
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Lead-paint completion certification present	Peeling paint flagged, no Form 442 completion cert in file	—
Lead-based paint disclosure present	Pre-1978 property, disclosure not in file	—
Self-employed P&L / balance sheet present	YTD P&L statement not in file for self-employed borrower	Self-Employed Income Documentation Index, p.1
No 30+ day mortgage lates in trailing 12mo	One 30-day late reported 04/22/2025; program requires zero	12-Month Mortgage Payment History (VOM), p.1
Appraisal not stale at closing	Appraisal effective date is 207 days before closing — exceeds the 120-day recertification limit	—

Loan 05 — 2025-1122-USDA-005 (USDA)

8 of 13 applicable checks failed — the most exceptions of the five, consistent with USDA's additional eligibility and ratio requirements.

Check	Finding	Source
Lead-paint completion certification present	Peeling paint flagged, no Form 442 completion cert in file	—
Lead-based paint disclosure present	Pre-1978 property, disclosure not in file	—
USDA property eligibility documented	Eligibility map screen-print/determination not in file	USDA Property Eligibility — Manual Review, p.1
Well & septic test documented	Private well + septic; inspections listed as "recommended," none in file	Appraisal Summary — USDA 502, p.1
Site value justification documented	Borderline site value ratio; USDA site value analysis not documented	—
USDA household income within moderate-income limit	\$134,720 adjusted household income exceeds the \$130,850 moderate-income limit	USDA GUS Findings, p.1
USDA PITI ratio within guideline	31.8% vs. 29% guideline; no waiver/compensating-factors documentation	USDA Debt Ratios & Waiver, p.1
USDA total debt ratio within guideline	43.9% vs. 41% guideline; no waiver/compensating-factors documentation	USDA Debt Ratios & Waiver, p.1

Methodology Notes

- **Determinism:** each loan's disposition is bound to its exact ruleset by a SHA-256 hash (recorded per loan in the underlying JSON export). Same loan + same ruleset hash will always reproduce the same result — no LLM runs at evaluation time.
- **NOT_APPLICABLE checks** (not itemized above) mean the relevant field simply has no value for that loan/program — e.g. USDA ratio checks are NOT_APPLICABLE on the VA and FHA loans. This is expected gating, not a data gap.
- **FLAG vs. FAIL:** a "FLAG" (as seen on Loan 02) means the closing document and the lender's system disagree on a value — informational only, since the document is treated as the source of truth and does not by itself force review.
- **Full audit trail:** every finding above traces to a specific document, page, and (where applicable) an exact text citation from the source file — available in the underlying `dispositions.json` export for anyone who needs to verify a specific line.